Radboud University Nijmegen

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The Prosciutto Question

Bachelor Thesis in Mathematics

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Contents

Chapter 1

Preliminaries

1.1 General knowledge and notation

We begin by recalling and defining some general tools that play an important role throughout this thesis. The first few concepts are often used to translate local properties into global results. This method plays a fundamental part in our later discussions about the formulation and analysis of weak forms and solutions to PDEs.

Let X be a topological space. A function $\phi : X \to \mathbb{R}$ is said to have *compact support* if the set

$$\operatorname{supp}(\phi) := \overline{\{x \in X \mid \phi(x) \neq 0\}}$$

is a compact subset of X. For any open $\Omega \subseteq \mathbb{R}^m$, the space of all smooth functions with compact support in Ω is denoted by $C_c^\infty(\Omega)$.

An important property of these functions is that they must be 0 on the boundary of Ω. In the particular context of PDE analysis, they are often referred to as *test functions*, but a more general term is *variations*. They have a wide range of applications, discussed in the field of the Calculus of Variations. A large portion of its theory is built around the following result, hence its name.

Lemma 1.1 (Fundamental Lemma of the Calculus of Variations)
Let $\Omega \subseteq \mathbb{R}^d$ be open and $f \in L^1_{\text{loc}}(\Omega)$. If

$$\int_\Omega f\phi = 0$$

holds for all $\phi \in C_c^\infty(\Omega)$, then $f = 0$ almost everywhere in Ω.

Multiplying with a test function and integrating is sometimes referred to as *testing against* a test function $\phi \in C_c^\infty(\Omega)$.

Definition 1.2
When we are differentiating functions in C^k defined in a subset of $\mathbb{R}^d$, we sometimes use the abbreviated notation

$$D^\alpha := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}},$$

where $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is called a *multi-index* and $|\alpha| := \alpha_1 + \dots + \alpha_d \leq k$.

Definition 1.3
Let $\Omega \subseteq \mathbb{R}^m$ be open. A function $f : \Omega \to \mathbb{R}$ is said to be *locally integrable* if

$$\int_K |f| < \infty$$

for all compact subsets K of Ω. The space of all locally integrable functions is denoted by $L^1_{\text{loc}}(\Omega)$, where we identify two functions with each other if they agree almost everywhere in Ω.

Put this before FLCV

An important fact that should be noted is that every function that is square-integrable is also locally integrable. That is, $L^2(\Omega) \subseteq L^1_{\text{loc}}(\Omega)$. For the sake of completeness, we will sometimes state definitions or results with the hypothesis that a function is locally integrable, only to use them on functions that are square-integrable.

1.2 Weak derivatives and Sobolev spaces

Before we talk about weak solutions to PDEs, we need to define spaces of functions that are potential candidates for solutions. We derive this space by weakening the conditions of differentiable functions.

Consider an open and bounded domain $\Omega \subseteq \mathbb{R}^d$. Then for any continuously differentiable function $u \in C^1(\Omega)$ and test function $\phi \in C_c^\infty(\Omega)$, we have

$$\int_\Omega u_{x_i} \phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_\Omega u \phi_{x_i} = - \int_\Omega u \phi_{x_i}, \tag{1.1}$$

since ϕ is 0 on the boundary. Consequentially, for any function $u \in C^k(\Omega)$ and multi-index $\alpha \in \mathbb{N}^d$ with $|\alpha| \leq k$, we have

$$\int_\Omega D^\alpha u \phi = (-1)^{|\alpha|} \int_\Omega u D^\alpha \phi, \tag{1.2}$$

using the fact that $D^\beta \phi$ has compact support as well, for every $\beta \in \mathbb{N}^d$ with $|\beta| \leq k$. Note that for this last expression to make sense, the function $D^\alpha u$ does not necessarily have to be continuous. In fact, we can completely remove the hypothesis that u is k times continuously differentiable, and define the *weak derivative* of u as a function that behaves in a similar way as Equation (1.2) when tested against a function $\phi \in C_c^\infty(\Omega)$.

Definition 1.4
Let $u, v \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{N}^d$. We say that v is the *α-th weak derivative* of u if

$$\int_\Omega v\phi = (-1)^{|\alpha|} \int_\Omega u D^\alpha \phi \tag{1.3}$$

for every test function $\phi \in C_c^\infty(\Omega)$.

We often use the same notation for weak derivatives as we do for regular derivatives. When $D^\alpha u$ is used, we should carefully think about which derivative is meant here. This notion extends to any differential operator that can classically be written in terms of partial derivatives, like the gradient ∇ or Laplacian $\Delta := \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_d^2}$. To be able to talk about *the* α-th weak derivative, we show that weak derivatives are unique.

Proposition 1.5
If the α-th weak derivative of u exists, it is unique up to a set of measure zero.

The proof of this proposition is a good example of how the Fundamental Lemma of the Calculus of Variations can be used.

Proof. Suppose that v and w are two α-th weak derivatives of u. Then for every $\phi \in C_c^\infty(\Omega)$,

$$\int_\Omega v\phi = (-1)^{|\alpha|} \int_\Omega u D^\alpha \phi = \int_\Omega w\phi,$$

which gives

$$\int_\Omega (v - w)\phi = 0$$

for every $\phi \in C_c^\infty(\Omega)$. We can now use the Fundamental Lemma of the Calculus of Variations (FLCV) to see that $v - w$ is 0 almost everywhere, which means that $v = w$ almost everywhere, completing the proof. $\square$

With this new notion of weak differentiability, we can define new spaces of regularity. These spaces are fundamental to the analysis of PDEs and will be used a lot in this thesis.

Definition 1.6
For $k \in \mathbb{N}$ and $p \in [1, \infty]$, the *Sobolev space of order k*, denoted $W^{k,p}(\Omega)$, is the space of functions for which the α-th weak derivative exists and is in $L^p(\Omega)$, for all $|\alpha| \leq k$:

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \mid D^\alpha u \in L^p(\Omega),\ |\alpha| \leq k\}.$$

When equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \le k} \|D^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p},$$

the space becomes complete, and thus forms a Banach space.

Recall that $L^p(\Omega)$ is a Banach space for every $p \in [1, \infty]$, but for $p = 2$ it also forms a Hilbert space. We can say this for Sobolev spaces as well. We write $H^k(\Omega) := W^{k,2}(\Omega)$ and equip it with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha| \le k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)},$$

forming a Hilbert space.

A useful subset of $W^{k,p}(\Omega)$ is $W_0^{k,p}(\Omega)$, which is defined as the closure of $C_c^\infty(\Omega)$ as a subset of $W^{k,p}(\Omega)$. For the case of $k = 2$, we write $H_0^k(\Omega) := W_0^{k,2}(\Omega)$.

Lastly, we define and $H^{-1}(\Omega)$ as the dual space of $H_0^1(\Omega)$, and denote the pairing between these two spaces as $\langle \cdot, \cdot \rangle$, i.e. $\langle f, u \rangle := f(u)$.

When $\partial\Omega$ is sufficiently smooth, we have a more intuitive way to interpret the space $H^1(\Omega)$. In fact, we can describe $H_0^1(\Omega)$ as the subset of $H^1(\Omega)$ containing all functions that vanish on the boundary. This interpretation has an important caveat. By definition, $H^1(\Omega)$ is a subset of $L^2(\Omega)$, so the functions that make up this Sobolev space are only defined up to a set of measure zero. The problem lies in the fact that the boundary $\partial\Omega$ is a set of measure zero, so we can generally not say anything about the values that are attained on it yet. We can, however, consider the case where we have a continuously differentiable function $u \in C^1(\overline{\Omega})$, for which the values on $\partial\Omega$ are certainly defined. We can also see u as a function in $H^1(\Omega)$, and define $v|_{\partial\Omega} := u|_{\partial\Omega}$ for every $v \in H^1(\Omega)$ that is equal to u almost everywhere in Ω. To further develop this idea, we first state the following property.

Proposition 1.7 (Approximation by smooth functions)

If the boundary $\partial\Omega$ is Lipschitz and $1 \le p < \infty$, the set of all smooth functions $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$.

More precisely, for any function $u \in W^{k,p}(\Omega)$, there is a sequence of smooth functions defined on the closure of Ω, $(u_n)_{n\in\mathbb{N}} \subseteq C^\infty(\overline{\Omega})$, such that

$$\|u - u_n\|_{W^{k,p}(\Omega)} \to 0 \qquad \text{as } n \to \infty.$$

This allows us to extend our knowledge of functions defined on the boundary to functions in $W^{k,p}(\Omega)$ in general.

Theorem 1.8 (Trace Theorem)
If the boundary $\partial\Omega$ is Lipschitz, there is a bounded linear operator $\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega)$ such that

(i) $\gamma u = u|_{\partial\Omega}$ for any $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$ and

(ii) $\|\gamma u\|_{L^2(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}$,

where the constant $C > 0$ only depends on p and Ω.

A full proof of this theorem can be found in [Eva10], but we will describe the general outline here.

Proof. As we previously discussed, we can start by defining $\gamma u := u|_{\partial\Omega}$ for any $u \in C^1(\overline{\Omega})$. The first step in extending this definition is to use that the boundary is Lipschitz to locally flatten it and prove item (ii) for $u \in C^1(\overline{\Omega})$ in a neighbourhood of some point $x \in \partial\Omega$. We can then use the fact that Ω is bounded to see that $\partial\Omega$ is compact, which allows us to get a finite amount of points $x_1, \dots, x_n \in \partial\Omega$ with corresponding constants $C_1, \dots, C_n$, combining them to conclude item (ii) for the entirety of $\partial\Omega$.

For the general case of $u \in W^{1,p}(\Omega)$, we can use Proposition 1.7 to get an approximation for u of smooth functions $(u_n)_{n\in\mathbb{N}} \subseteq C^\infty(\overline{\Omega})$. The first part of the proof then implies that

$$\|\gamma u_n - \gamma u_m\|_{L^2(\partial\Omega)} = \|\gamma (u_n - u_m)\|_{L^2(\partial\Omega)} \leq C \|u_n - u_m\|_{W^{k,p}(\Omega)},$$

which means that $(\gamma u_n)_{n\in\mathbb{N}}$ is a Cauchy sequence. We have used that γ is linear on $C^1(\overline{\Omega})$, since it simply acts as a restriction there. Using the fact that $L^2(\partial\Omega)$ is a Hilbert space, we can define

$$\gamma u := \lim_{n\to\infty} \gamma u_n = \lim_{n\to\infty} u_n|_{\partial\Omega},$$

which satisfies the same bound.

The proof is completed by noting that if $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$, then there are smooth functions $(u_n)_{n\in\mathbb{N}} \subseteq C^\infty(\overline{\Omega})$ converging to u uniformly on $\overline{\Omega}$, so $\gamma u = u|_{\partial\Omega}$. □

Definition 1.9
This operator $\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega)$ is referred to as the *trace operator*. Then γu is called the *trace* of u, and is sometimes denoted by $u|_{\partial\Omega}$.

1.3 Weak time derivatives

In the field of PDEs, the requirements on time regularity and space regularity are often different. Let for example

Let X be a Banach space and $T > 0$. A function $f : [0, T] \to X$ is called *strongly measurable* if there is a sequence of simple functions $(f_n)_{n\in\mathbb{N}}$ that converges pointwise to f at almost every

$t \in [0, T]$. Now for $p \in [1, \infty]$, the space $L^p(0, T; X)$ consists of all strongly measurable functions u such that

$$\|u\|_{L^p(0,T;X)} := \left(\int_0^T \|u(t)\|_X \, dt \right)^{1/p} < \infty$$

for $p \in [1, \infty)$, and

$$\|u\|_{L^\infty(0,T;X)} := \operatorname*{ess\,sup}_{t \,\in\, [0,T]} \|u(t)\|_X < \infty.$$

Lastly, $C([0, T]; X)$ denotes the space of all continuous functions $u : [0, T] \to X$ with

$$\|u\|_{C([0,T];X)} := \max_{t \in [0,T]} \|u(t)\|_X < \infty.$$

We can take X to be, for instance, $L^2(\Omega)$. Then formally, $C([0, T]; L^2(\Omega))$ consists of continuous functions u from the interval time [0, T] to $L^2(\Omega)$. For a more intuitive interpretation, you might feel inclined to interpret them as the corresponding functions $\widetilde{u}$ from $[0, T] \times \Omega$ to $\mathbb{R}$, given by $\widetilde{u}(t, x) = u(t)(x)$. We do need to be careful when we use this interpretation, though, as the notation might make it seem like these functions $\widetilde{u}$ are continuous in time, while this is generally not the case. A more accurate way to think about this, is that at every time t, the set of points $x \in \Omega$ where $\widetilde{u}$ has a discontinuity in time has measure zero.

Caratheodory ?

1.4 Variational forms

Using the same method, we can weaken the requirements for solutions to a PDE. Consider, for instance, the heat equation on $\Omega_T := (0, T) \times \Omega$ with homogeneous Dirichlet boundary conditions. For a function u to be a classical solution, we require u to be of class $C^{1,2}(\Omega_T) \cap C^0(\overline{\Omega_T})$, i.e. continuously differentiable with respect to t, twice continuously differentiable with respect to x and continuously extendable to $\overline{\Omega_T}$. We can multiply the heat equation by a test function $\phi \in C_c^\infty(\Omega)$ and integrate on Ω to see

rewrite

$$\int_\Omega u_t \phi = \int_\Omega \Delta u \phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_\Omega \nabla u \cdot \nabla \phi = - \int_\Omega \nabla u \cdot \nabla \phi. \tag{1.4}$$

We can also write this as

$$\langle u_t, \phi \rangle = -\langle \nabla u, \nabla \phi \rangle, \tag{1.5}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product on both $L^2(\Omega)$ and $L^2(\Omega; \mathbb{R}^m)$.

Chapter 2

Turing patterns

Chapter 3

A bulk-surface model

In this chapter, we will discuss a bulk-surface model that is motivated by the interactions of ligands and receptors. We will give a classical proof of existence and uniqueness of weak solutions to this model, as well as discuss some of the qualitative properties of such solutions. The proof will rely heavily on the definitions and results introduced in chapter 1, and will we based on an application of the Banach fixed-point theorem.

3.1 Biological motivation

actual biological terms

Consider an open and bounded set $U \subseteq \mathbb{R}^d$ with Lipschitz boundary and $T > 0$. This can be interpreted as the microscopic part of the body we are investigating. We identify the cells in this part of the body by an open set $C \subseteq U$, also with Lipschitz boundary, and the extracellular space by $\Omega := U \setminus \overline{C}$. We require that the closure $\overline{C}$ of C is also completely contained by U, i.e. no cells should cross the boundary of our domain. In particular, this implies that the boundary $\partial\Omega$ of the extracellular space can be written as the disjoint union of the boundary of our region of investigation ∂U and the cell membranes ∂C, which we denote by Γ_e and Γ, respectively. We use the function $u : [0,T] \times \Gamma \to \mathbb{R}$ to represent the concentration of free receptors at a given time and place on a cell membrane. Similarly, $u_b : [0,T] \times \Gamma \to \mathbb{R}$ represents the concentration of bounded receptors, and $v : [0,T] \times \overline{\Omega} \to \mathbb{R}$ represents the concentration of ligands. figure

The ligands can move freely through the extracellular space. This can modelled by the standard diffusion (or heat) equation. For the sake of simplification, we assume the individual receptors will be fixed in their position on the cell membrane.

When a ligand binds to a free receptor, the concentration of free receptors and ligands decrease, and the concentration of bounded receptors at that position increases accordingly. The rate of this reaction is proportional to $u^2 v$. or uv After some time, the bounded receptor has done its task, passing the signal through to the interior of the cell. It releases the ligand and is now available to bind to a new one again. The concentration of bounded receptors decrease, and the concentration of free receptors and ligands increase accordingly. The rate of this dissociation is proportional to the concentration of bounded receptors u_b.

The last two things that happen constantly are production and decay. To keep intercellular communication possible, cells produce ligands and receptors that enter or model through the cell

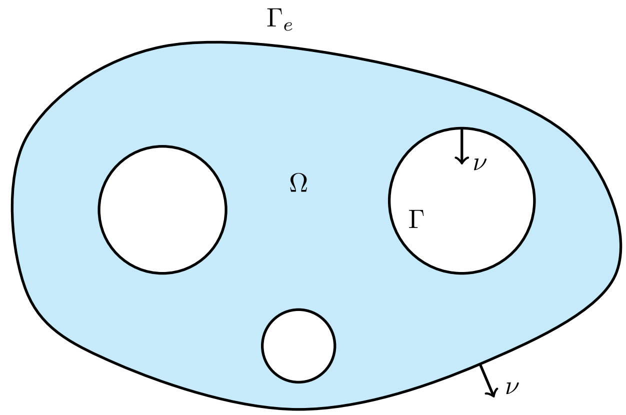

Figure 3.1: A sketch of a possible domain in the case that $d = 2$. The blue background represents Ω and the two kinds of boundaries are indicated, along with the outward unit normal vector ν.

membrane. We assume this production happens at a constant rate. The free and bounded receptors also decay after some time, proportional to their respective concentrations.

We can describe this behaviour with a system of coupled differential equations. Denote the association and dissociation coefficients by $k_{\text{on}}, k_{\text{off}} > 0$, respectively, the production constants by $p_u, p_v > 0$, the decay coefficients by $d_u, d_{u_b} > 0$ and the diffusion coefficient by $D > 0$. Furthermore, we denote the outward unit normal vector on $\partial\Omega$ by ν. Then the interactions between ligands and receptors are described by

$$f(u, u_b, v) := k_{\text{on}} u^2 v - k_{\text{off}} u_b,$$

and the production and decay of u, u_b and v are described by

$$p(u) := p_u - d_u u, \qquad q(u_b) := -d_{u_b} u_b \qquad \text{and} \qquad r(v) := p_v,$$

respectively. The concentrations of receptors can then be described by the system of ODEs given by

$$\begin{aligned} u_t &= -f(u, u_b, \gamma v) + p(u) \quad &&\text{on } (0, T) \times \Gamma, \\ u_{b,t} &= f(u, u_b, \gamma v) + q(u_b) \quad &&\text{on } (0, T) \times \Gamma, \end{aligned} \tag{3.1}$$

where γv denotes the trace of v, as discussed in section 1.2. Note that even though they are given by only two equations, this is actually a system of uncountably many ODEs. For every boundary point $x \in \Gamma$, we have one ODE for the function $t \mapsto u(t, x)$ and one for the function $t \mapsto u_b(t, x)$.

Since the dynamics of the ligands include diffusion, they can be described by the PDEs given by

$$\begin{aligned} v_t &= D\Delta v \quad &&\text{in } (0, T) \times \Omega, \\ D\frac{\partial v}{\partial \nu} &= -f(u, u_b, v) + r(v) \quad &&\text{on } [0, T) \times \Gamma, \\ D\frac{\partial v}{\partial \nu} &= 0 \quad &&\text{on } [0, T) \times \Gamma_e. \end{aligned} \tag{3.2}$$

Additionally, these concentrations can be required to adhere to some initial conditions. That is, they should satisfy

$$\begin{aligned} u(0,\ \cdot\ ) &= u_0 &&\text{on } \Gamma,\\ u_b(0,\ \cdot\ ) &= u_{b,0} &&\text{on } \Gamma,\\ v(0,\ \cdot\ ) &= v_0 &&\text{on } \Omega, \end{aligned} \tag{3.3}$$

for some $u_0, u_{b,0} : \Gamma \to \mathbb{R}$ and $v_0 : \Omega \to \mathbb{R}$.

The remainder of this chapter will be dedicated to studying the system of Equations (3.1) to (3.3).

3.2 Weak formulation

Before we start proving the existence of solutions to this model, we must discuss what kind of solutions we are looking for. Specifically, we will derive the weak formulations of Equations (3.1) to (3.3) and discuss the function spaces that are involved.

As in section 1.4, we first suppose that u, u_b and v are smooth solutions. Then we can interpret them as functions $\mathbf{u} : [0,T] \to L^2(\Gamma)$, $\mathbf{u_b} : [0,T] \to L^2(\Gamma)$ and $\mathbf{v} : [0,T] \to H^1(\Omega)$, defined by

$$\mathbf{u}(t) := u(t,\cdot), \qquad \mathbf{u_b}(t) := u_b(t,\cdot) \qquad \text{and} \qquad \mathbf{v}(t) := v(t,\cdot).$$

Now if we denote the weak time derivatives of $\mathbf{u}$ and $\mathbf{u_b}$ by $\mathbf{u}'$ and $\mathbf{u_b}'$, respectively, we can write Equation (3.1) as

$$\begin{aligned} \mathbf{u}' &= -f(\mathbf{u},\mathbf{u_b},\mathbf{v}) + p(\mathbf{u}),\\ \mathbf{u_b}' &= f(\mathbf{u},\mathbf{u_b},\mathbf{v}) + q(\mathbf{u_b}). \end{aligned} \tag{3.4}$$

Weak solutions of Equation (3.1) are then required to satisfy Equation (3.4) for almost every $t \in [0,T]$ and $x \in \Omega$. or integral representation

Contrary to the ODEs, the PDE is not defined separately for every x. Instead, we have to work on the entire domain simultaneously. By multiplying $\mathbf{v}'$ by a test function $\phi \in C_c^\infty(\Omega)$ and integrating on Ω for fixed t, we get

$$\begin{aligned} \int_\Omega \mathbf{v}'\phi = D\int_\Omega \Delta\mathbf{v}\phi &= -D\int_\Omega \nabla\mathbf{v}\cdot\nabla\phi + D\int_{\partial\Omega}\frac{\partial\mathbf{v}}{\partial\nu}\phi\, dS\\ &= -D\int_\Omega \nabla\mathbf{v}\cdot\nabla\phi + D\int_{\Gamma}\frac{\partial\mathbf{v}}{\partial\nu}\phi\, dS + D\int_{\Gamma_e}\frac{\partial\mathbf{v}}{\partial\nu}\phi\, dS\\ &= -D\int_\Omega \nabla\mathbf{v}\cdot\nabla\phi - \int_\Gamma f(\mathbf{u},\mathbf{u_b},\mathbf{v})\phi\, dS + \int_\Gamma r(\mathbf{v})\phi\, dS, \end{aligned}$$

which we can also write as

$$(\mathbf{v}',\phi)_{L^2(\Omega)} = -D\,(\nabla\mathbf{v},\nabla\phi)_{L^2(\Omega;\mathbb{R}^d)} - (f(\mathbf{u},\mathbf{u_b},\mathbf{v}),\phi)_{L^2(\Gamma)} + (r(\mathbf{v}),\phi)_{L^2(\Gamma)}\,. \tag{3.5}$$

Since we have used integration with respect to space in this computation, this expression only depends on time. A weak solution of Equation (3.2) is therefore required to satisfy Equation (3.5) at almost every $t \in [0,T]$. define $\mathcal{X}$ and $\mathcal{Y}$

Definition 3.1
The functions $\mathbf{u}, \mathbf{u_b} \in \mathcal{X}$ and $\mathbf{v} \in \mathcal{Y}$ are *weak solutions* to Equations (3.1) to (3.3) if they satisfy Equation (3.4) for almost every $t \in [0, T]$ and $x \in \Omega$, and Equation (3.5) for almost every $t \in [0, T]$.

correct?

3.3 Existence and uniqueness

Our goal in this section is to prove the following theorem.

Theorem 3.2 (Existence and uniqueness of weak solutions)
Let $u_0, u_{b,0} \in L^2(\partial\Omega)$ and $v_0 \in L^2(\Omega)$ be non-negative initial conditions. Then there exist unique weak solutions $u, u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ to Equations (3.1) to (3.3). Moreover, these solutions are also non-negative.

As previously mentioned, the proof will be structured around an application of the Banach fixed-point theorem. If you have followed a course on ODE theory, chances are that you have seen the Picard-Lindelöf Theorem and its proof. The proof for our model follows the same general idea, but requires some extra steps to handle our situation of a coupled system. We "freeze" the terms in the ODEs that depend on v so we can use the standard existence and uniqueness properties we know about such systems to get solutions u and u_b to this alternative problem. The PDE gets a similar treatment, producing solutions v of some alternative PDE, after which we can "update" the frozen terms in the ODE. This produces an operator $\mathcal{T}$ that maps any $v \in \mathcal{Y}$ to another $\widetilde{v} \in \mathcal{Y}$. If we can then show that this operator is a contraction, the Banach fixed-point theorem will guarantee the existence and uniqueness of a fixed point $v^* \in \mathcal{Y}$, which will be our solution to the PDE.

Let us first concretely demonstrate what we mean by "freezing" the terms that depend on v. Fix $v^* \in \mathcal{Y}$ and consider the modified system of ODEs given by

$$\begin{aligned} u_t &= -f(u, u_b, \gamma v^*) + p(u) \quad \text{on } (0,T) \times \Gamma, \\ u_{b,t} &= f(u, u_b, \gamma v^*) + q(u_b) \quad \text{on } (0,T) \times \Gamma. \end{aligned} \tag{3.6}$$

weak form?

These ODEs are now *decoupled* from the PDE, so we can freely talk about the existence and uniqueness of solutions to this system, without simultaneously having to worry about the PDE.

Lemma 3.3 (Existence and uniqueness for the decoupled ODEs)
Let $u_0, u_{b,0} \in L^2(\partial\Omega)$ be non-negative initial conditions and fix $v^* \in \mathcal{Y}$. Then there exist unique weak solutions $u, u_b \in \mathcal{X}$ to Equation (3.6) satisfying these initial conditions. Moreover, these solutions are also non-negative and bounded by a constant depending on T.

Proof. Let us begin by summarizing the problem in one formula. Fix $x \in \Omega$ and denote

$$w(t) := \begin{pmatrix} u(t,x) \\ u_b(t,x) \end{pmatrix}.$$

Then Equation (3.6) can be described by

$$w'(t) = F(w(t), t), \tag{3.7}$$

where $F : \mathbb{R}^3 \to \mathbb{R}^2$ is defined by

$$F(w,t) :=$$

□

Using Lemma 3.3, we define an operator $\mathcal{T}_1 : \mathcal{Y} \to \mathcal{X}^2$ that maps any function $v^* \in \mathcal{Y}$ to a tuple $(u, u_b) \in \mathcal{X}^2$ of solutions to Equation (3.6).

We can use a similar method for the PDE in Equation (3.2). Fix $(u^*, u_b^*) \in \mathcal{X}^2$ and consider the modified weak formulation given by

$$\begin{aligned}(v_t, \phi)_{L^2(\Omega)} &= -\left(f(u^*, u_b^*, v), \phi\right)_{L^2(\Gamma)} + (r(v), \phi)_{L^2(\Gamma)} - D\left(\nabla v, \nabla \phi\right)_{L^2(\Omega;\mathbb{R}^d)} \\ &= (g(v), \phi)_{L^2(\Gamma)} - D\left(\nabla v, \nabla \phi\right)_{L^2(\Omega;\mathbb{R}^d)},\end{aligned} \tag{3.8}$$

fix more?

where $g(v) := -f(u^*, u_b^*, v) + r(v)$. Note that we can only make this definition because u^* and u_b^* are fixed. We have now decoupled the PDE from the ODEs, and we can prove a result that is nearly identical to Lemma 3.3.

Lemma 3.4 (Existence and uniqueness for the decoupled PDE)
Let $v_0 \in L^2(\Omega)$ be non-negative initial conditions and fix $(u^*, u_b^*) \in \mathcal{X}^2$. Then there exists a unique weak solution $v \in \mathcal{Y}$ to Equation (3.8) satisfying these initial conditions. Moreover, this solution is also non-negative and bounded by a constant depending on T.

Proof. □ proof

This result allows us to define another operator $\mathcal{T}_2 : \mathcal{X}^2 \to \mathcal{Y}$ sending a tuple $(u^*, u_b^*) \in \mathcal{X}^2$ to a solution $v \in \mathcal{Y}$ of Equation (3.8). By combining this step with the previous one, we can use $\mathcal{T} := \mathcal{T}_2 \circ \mathcal{T}_1 : \mathcal{Y} \to \mathcal{Y}$ to iteratively get new solutions. At each step, we modify Equation (3.6) using the input from $\mathcal{Y}$ to get a new pair $(u, u_b) \in \mathcal{X}^2$, with which we proceed to update Equation (3.8) to get a new $v \in \mathcal{Y}$.

Lemma 3.5
For $\widetilde{T} > 0$ small enough, $\mathcal{T} : \mathcal{Y} \to \mathcal{Y}$ is a contraction. That is, there is a $q \in [0, 1)$ such that

$$\|\mathcal{T} v\|_{\mathcal{Y}} \leq q \, \|v\|_{\mathcal{Y}}$$

for every $v \in \mathcal{Y}$.

Proof. □ proof

We have now gathered enough results to prove the main statement of this section.

Proof of Theorem 3.2. Using Lemma 3.5, let $T > 0$ such that $\mathcal{T}$ is a contraction. Then by the Banach fixed-point theorem, there exists a unique fixed point v^* of $\mathcal{T}$. Next, define $(u^*, u_b^*) := \mathcal{T}_1 v^*$. Then by definition, u^* and u_b^* solve Equation (3.6) with v^* as the frozen input. On the other hand, $\mathcal{T}_2(u^*, u_b^*) = \mathcal{T} v^* = v^*$ solves Equation (3.8) with u^* and u_b^* as the inputs to the frozen reactions. In other words, u^*, u_b^* and v^* are weak solutions to Equations (3.1) to (3.3).

To show uniqueness of the weak solutions, it is enough to note that for any weak solution u^*, u_b^* and v^*, we have that v^* is a fixed point of $\mathcal{T}$. This follows directly from the uniqueness results from Lemmas 3.3 and 3.4. Since the Banach fixed-point theorem tells us that fixed points are unique, we know that weak solutions to Equations (3.1) to (3.3) are too. □

Chapter 4

Simulations

4.1 Problem definitions

4.1.1 Non-diffusing receptors on the cell boundary

This problem assumes the receptor lives on the cell boundaries without diffusing. The ligand is able to diffuse across the entire extracellular space, and interacts with the receptor at the cell boundaries according to the relevant ordinary differential equation. Production of both ligand and receptor exclusively takes place on the cell boundaries.

Let $\Omega \subseteq \mathbb{R}^m$ be the open and bounded domain and denote the cells by an open set $C \subseteq \Omega$ of which the closure $\overline{C}$ is contained in Ω. Furthermore, let $E = \Omega \setminus \overline{C}$ be the extracellular space, and let $l(t, x)$ denote the concentration of ligands at time $t > 0$ and point $x \in \overline{E}$, and $r(t, x)$ the concentration of receptors at time $t > 0$ and point $x \in \partial C$. Then the unknowns l and r should satisfy

$$\begin{aligned} r_t &= \gamma(a - r + r^2 l) && \text{on } (0,\infty) \times \partial C, \\ l_t &= d(???)\Delta l && \text{in } (0,\infty) \times E, \\ \frac{\partial l}{\partial \nu} &= \gamma(b - r^2 l) && \text{on } (0,\infty) \times \partial C, \\ \frac{\partial l}{\partial \nu} &= 0 && \text{on } (0,\infty) \times \partial\Omega, \end{aligned} \tag{4.1}$$

where $d > 0$ is the diffusion coefficient, $\gamma > 0$ is the reaction coefficient and $a > 0$ and $b > 0$ are the production coefficients of the receptor and ligand respectively.

Let $\tau > 0$ be the time discretization parameter and for every $n \in \mathbb{N}$, let $t_n := n\tau$ and

$$r_n(x) \approx r(t_n, x) = r(n\tau, x) \quad \text{and} \quad l_n(x) \approx l(t_n, x) = l(n\tau, x) \tag{4.2}$$

be the approximations of receptor and ligand at time $t_n = n\tau$ respectively. Since we only consider an ODE for r, we do not need to formulate a variational form for this unknown. Instead, we will approximate the ODE for r with some finite difference quotient (???),

$$\frac{r_{n+1} - r_n}{\tau} = \gamma(a - r_n + r_n^2 l_n) \quad \text{on } \partial C, n \in \mathbb{N}, \tag{4.3}$$

using forward Euler. This gives us an explicit formula for r_{n+1},

$$r_{n+1} = r_n + \tau\gamma\left(a - r_n + r_n^2 l_n\right) \quad \text{on } \partial C, n \in \mathbb{N}, \tag{4.4}$$

with r_0 initialized as the exact discretization of the initial conditions on r. To approximate l, we first discretize time using a method slightly different as the one used for r,

$$\frac{l_{n+1} - l_n}{\tau} = d\Delta l_{n+1} \quad \text{in } E, n \in \mathbb{N}. \tag{4.5}$$

The use of the backward Euler instead of forward Euler method provides additional stability, which is particularly convenient considering we are dealing with a second-order derivative. This gives us the implicit formula for l_{n+1},

$$l_{n+1} - \tau d\Delta l_{n+1} = l_n \quad \text{in } E, n \in \mathbb{N}. \tag{4.6}$$

As for r, l_0 is initialized as the exact discretization of the initial condition on l. To derive the variational form of this problem, we multiply both sides by a test function ϕ, integrate on E and use partial integration,

$$\begin{aligned}\int_E l_n\phi &= \int_E \left[l_{n+1}\phi - \tau d\Delta l_{n+1}\phi\right] \\ &= \int_E l_{n+1}\phi + \tau d\int_E \nabla l_{n+1}\nabla\phi - \tau d\int_{\partial E}\frac{\partial l_{n+1}}{\partial\nu}\phi\, dS \\ &= \int_E l_{n+1}\phi + \tau d\int_E \nabla l_{n+1}\nabla\phi - \tau d\int_{\partial C}\gamma(b - r_{n+1}^2 l_{n+1})\phi\, dS,\end{aligned} \tag{4.7}$$

where in the last step we used that ∂E is the disjoint union of $\partial\Omega$ and ∂C, since $\overline{C} \subseteq \Omega$, and the Neumann boundary conditions.

$$\int_{\partial C} r_{n+1}\phi = \int_{\partial C}\left[r_n\phi + \tau\gamma\left(a - r_n + r_n^2 l_n\right)\phi\right] \tag{4.8}$$

Chapter 5

Bibliography

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